

Homework 4

1. See Figure 1 below. While values of epsilon lower than 0.1 showed significant exponential decay in error, a higher epsilon value of 0.4, meaning the robot has a 40% chance to make an incorrect observation, results in a high initial localization error that does not significantly decrease over time.

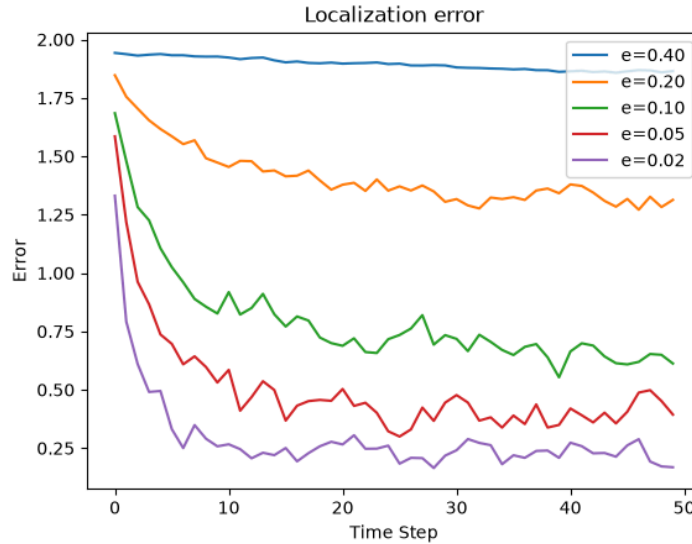


Figure 1.

2. See Figure 2. The localization error for particle filtering tends to have both a higher initial value and a lower decay rate, suggesting a worse overall performance. This may be because particle filtering has only a default of 30 particles to approximate the probability distribution of the agent's location, in comparison to the N by T belief state that exact inference has access to.

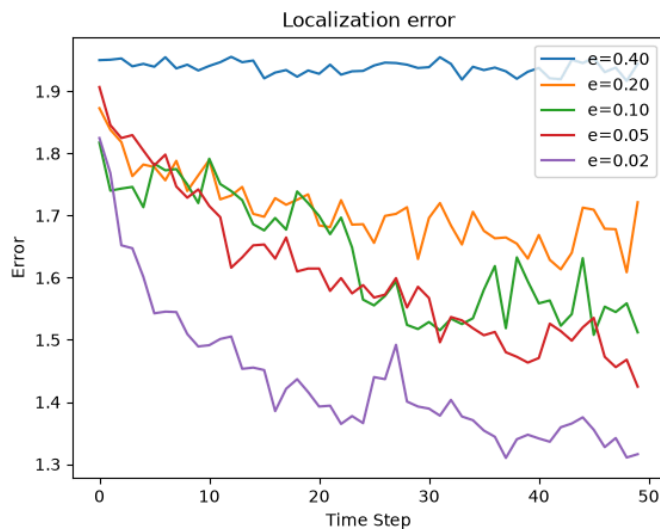


Figure 2.

3. Errors occur when the particle filtering algorithm predicts an incorrect starting location for the agent. Even as the algorithm remains confident about the agent's location, it can only resample from the existing, incorrect distribution, making it unlikely for the algorithm to recover the robot's true position.